Whitepaper

ESP32 RV Monitoring Gateway

A Modular IoT Platform for Recreational Vehicles and Vessels

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Introduction

Recreational vehicles and boats are increasingly equipped with complex electrical and electronic systems. The status of these systems can typically be accessed locally through various mobile applications, displays, or proprietary interfaces. As long as the owner is near the vehicle, important information such as battery voltage, state of charge, temperature, and system alerts is readily available.

However, when a recreational vehicle or boat is left unattended for an extended period, access to this information is often limited or unavailable. In precisely these situations, remote monitoring can be highly valuable, enabling owners to track battery status, environmental conditions, and potential system faults before they become critical issues.

The RV Monitoring Gateway aims to provide location-independent access to selected operational and status data from recreational vehicles and boats through an internet connection. The system is designed to offer a simple, modular, and extensible platform for remotely monitoring critical onboard systems, improving both convenience and peace of mind for vehicle owners

Motivation

Existing monitoring solutions range from simple device-specific mobile applications to comprehensive monitoring platforms capable of integrating entire onboard electrical systems. However, these solutions often depend on equipment that must remain powered continuously or operate in a standby mode to provide remote access.

For vessels and recreational vehicles that are left unattended for extended periods, maintaining large parts of the electrical system in an active state may be undesirable due to power consumption, system complexity, or reliability concerns. Owners may nevertheless wish to monitor a small set of critical parameters such as battery state of charge, cabin temperature, humidity levels, or vessel location.

In addition, many existing solutions require extensive integration into the onboard infrastructure, making retrofitting costly and time-consuming, particularly on older vessels and vehicles.

These challenges highlight the need for a lightweight monitoring gateway that can operate with minimal power consumption and installation effort while providing access to essential information from anywhere with an internet connection. The system is intended to provide its core functionality independently of existing onboard monitoring ecosystems while remaining open to future integration with third-party devices and services.

Requirements

The RV Monitoring Gateway is intended to provide remote access to essential operational data of recreational vehicles and boats while minimizing installation effort, power consumption, and dependency on proprietary ecosystems.

Based on these objectives, the following key requirements were identified:

Remote Accessibility

The system shall provide remote access to relevant monitoring information from any location with internet connectivity. Users shall be able to review the status of their vehicle or vessel while it is unattended, regardless of their physical location.

The specific implementation of the communication infrastructure is not defined by this requirement.

Low Power Consumption

The monitoring gateway may be powered either from the vehicle's existing battery system or from an independent auxiliary power supply, depending on the installation requirements.

As recreational vehicles and vessels may remain unattended for extended periods without access to external power, the system shall be optimized for minimal energy consumption.

The gateway therefore operates primarily in a low-power mode, with the ESP32 remaining in deep sleep between periodic sensor acquisition cycles. Communication modules are activated only when telemetry data or alarm notifications need to be transmitted. This design enables continuous monitoring even when major onboard systems are switched off, while minimizing battery drain and maximizing operating autonomy.

Non-Invasive Installation

The RV Monitoring Gateway shall be designed as a retrofit solution requiring minimal intervention in the existing electrical system.

Essential monitoring functions shall be achievable using sensors directly connected to the gateway. Where available, data from existing onboard sensors, monitoring devices, and communication interfaces may be integrated to extend the available information.

This concept enables reliable operation with minimal installation effort and without dependence on specific onboard equipment.

Vendor Independence

The RV Monitoring Gateway shall provide its essential monitoring functions using sensors directly connected to the gateway and shall not require a specific hardware manufacturer or device ecosystem.

Where available, additional data may be obtained from vendor-specific devices through suitable communication interfaces or APIs. These integrations are optional and intended to extend the available information rather than provide core functionality.

Reliable Communication

The RV Monitoring Gateway shall support different communication technologies depending on the available infrastructure. While the initial prototype focuses on Wi-Fi connectivity, the architecture is designed to support cellular communication in future versions.

To ensure reliable operation, the gateway shall be capable of temporarily storing monitoring data when no communication link is available. Buffered data should be transmitted automatically once connectivity has been restored.

Appropriate security measures shall be implemented to protect the confidentiality and integrity of transmitted and stored data. These measures may include encrypted communication channels, authentication mechanisms, and controlled access to monitoring information.

Expandability

The system architecture shall be designed to allow future extensions beyond the initial monitoring functions. This includes support for additional sensors, communication interfaces, monitoring capabilities, and data processing functions.

These features are not required for the initial prototype but shall be considered when designing the hardware interfaces, software structure, and communication concept.

System Concept

Architecture

The RV Monitoring Gateway follows a layered architecture consisting of four primary components: data acquisition, gateway processing, backend services, and user access.

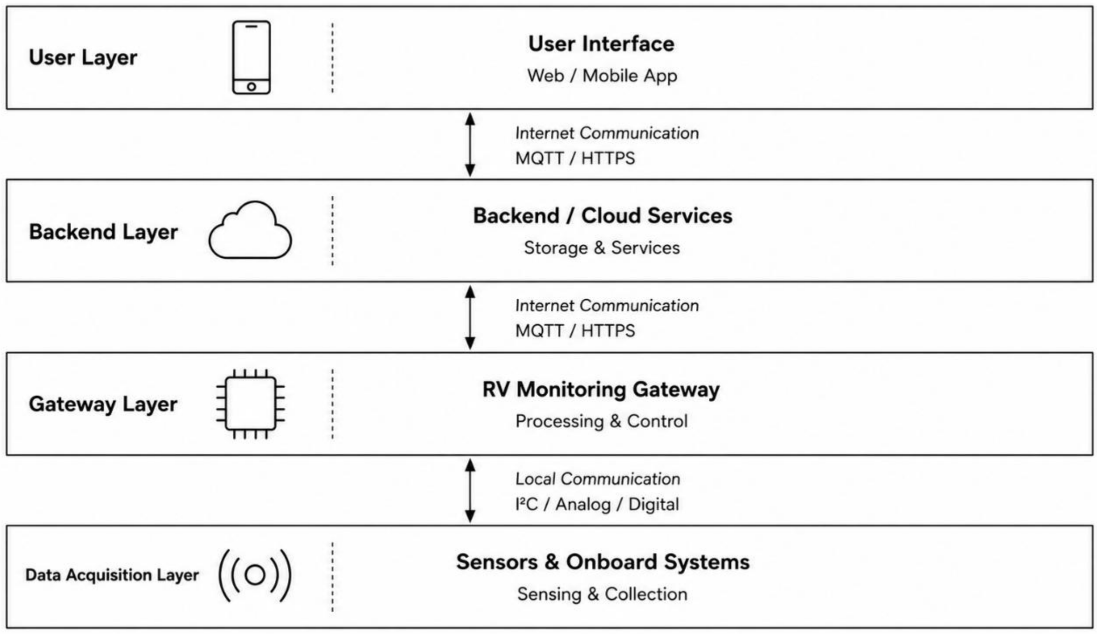
The data acquisition layer is responsible for collecting information from local or onboard sensors and, where available, existing monitoring systems. This layer provides the raw data required for condition monitoring.

The gateway layer serves as the central processing. It acquires sensor data, performs local processing and validation, manages temporary data storage, and prepares information for transmission.

The backend layer provides centralized services such as data storage, user authentication, historical data management, and remote access capabilities. It acts as an intermediary between the gateway and end users.

The user access layer presents monitoring information through a web or mobile interface, allowing users to review vehicle status and receive notifications regardless of their physical location.

Communication between the individual layers is implemented using suitable communication technologies and may vary depending on the available infrastructure and deployment scenario.



This separation of responsibilities simplifies future system extensions and allows individual components to evolve independently.

Data Sources

The initial prototype focuses on monitoring a limited set of essential parameters relevant to recreational vehicles and vessels.

The following data sources are currently considered, as they monitor conditions that may require user attention while the vehicle is unattended.

Parameter	**Source**	**Interface**
Engine battery voltage	Direct measurement	Analog Input
House battery voltage	Direct measurement	Analog Input
Gateway supply voltage	Direct measurement	Analog Input
Temperature	Environmental sensor	I²C
Humidity	Environmental sensor	I²C
Water ingress detector	Switch contact	Digital Input
Smoke alarm	Relays contact	Digital Input

Additional data sources may be integrated in future versions, including GPS position, tank levels, solar charging information, and data from onboard monitoring systems.

Communication

The communication concept of the RV Monitoring Gateway distinguishes between **regular telemetry data** and **event-based alarm messages**.

Regular telemetry data shall be acquired periodically by the gateway, for example every five minutes. This data shall be transmitted to a backend service where it is stored in a database. The stored data can then be accessed through a user interface and displayed either as current status information or as historical trends.

To verify that the gateway remains operational, the backend shall evaluate the regular telemetry messages received from the device. Each telemetry message acts as an **implicit heartbeat** by confirming that the gateway is powered, running, and able to communicate.

If no telemetry message is received within a defined time window, the backend may classify the gateway as offline and trigger a notification to the user. This approach avoids the need for a separate heartbeat mechanism on the gateway while still allowing the system to detect a loss of monitoring capability.

In addition to periodic telemetry transmission, the gateway shall support event-based communication for critical conditions. Events such as smoke detection, water ingress, or critically low battery voltage should trigger an immediate alarm message rather than waiting for the next scheduled transmission cycle.

Alarm messages shall be generated by the gateway and transmitted with minimal delay. Depending on the available communication infrastructure and system configuration, alarm notifications may be routed through the backend service, transmitted via external notification services, or delivered directly by the gateway hardware.

This separation between periodic telemetry and event-based alarms improves both efficiency and responsiveness. Normal operating data can be transmitted at a low frequency to reduce power consumption and network usage, while critical events can still be reported with minimal delay.

Hardware

The initial prototype is based on a low-power microcontroller platform that combines data acquisition, local processing, and communication capabilities within a single device. The selected hardware is intended to provide a balance between functionality, power consumption, cost, and ease of integration.

The gateway hardware consists of three primary subsystems:

- Data acquisition
- Processing Unit
- Communication Interfaces

Data acquisition is performed either through dedicated monitoring sensors connected directly to the gateway or through data obtained from existing onboard equipment via suitable communication interfaces. This approach enables the gateway to provide independent monitoring capabilities while also utilizing available information from existing monitoring and control systems.

The **processing unit** serves as the central component of the RV Monitoring Gateway. It is responsible for acquiring data from connected sensors and communication interfaces, performing local data processing, managing temporary data storage, and coordinating communication with backend services.

In addition to regular telemetry handling, the processing unit evaluates incoming sensor data for critical conditions such as water ingress, smoke detection, or low battery voltage. When such events are detected, the processing unit generates alarm messages and initiates the appropriate notification mechanism.

To support operation in environments with intermittent network connectivity, the processing unit may temporarily buffer monitoring data and transmit it once communication has been restored. It therefore acts as the central coordination and decision-making element of the monitoring system.

Communication interfaces provide connectivity to external services. The initial prototype focuses on Wi-Fi communication, while the architecture allows future integration of cellular communication modules for operation in locations without local network access.

The majority of the gateway functionality can be implemented using a small number of hardware components. The following table summarizes the primary hardware components and their responsibilities within the system architecture.

Subsystem	Hardware	Purpose
Data Acquisition	SHT31 / DHT22	Temperature and humidity monitoring
Processing, Data Acquisition and Communication	ESP32	Voltage measurement, water and smoke alarm monitoring, local data processing, alarm generation, data buffering, Wi-Fi and BLE communication
Communication	SIM7000E (future)	Cellular communication and remote connectivity

Services and Backend Infrastructure

The monitoring gateway is designed to support integration with backend services for data storage, visualization, device management, remote configuration, and alarm handling. However, the specific backend architecture is outside the scope of this whitepaper and will be selected during the implementation phase based on project requirements.

The backend implementation is intentionally technology-agnostic and may be realized using self-hosted, cloud-based, or hybrid infrastructure. The final solution shall be chosen according to factors such as scalability, operational costs, security requirements, connectivity constraints, and customer preferences.

This approach ensures maximum flexibility while avoiding unnecessary restrictions on future system development and deployment strategies.

Security and Reliability

The RV Monitoring Gateway is intended to operate unattended for extended periods and therefore requires mechanisms to ensure both reliable operation and secure access to monitoring information.

Reliability

The system shall be designed for reliable operation in recreational vehicles and vessels where power and communication infrastructure may not always be continuously available. Temporary interruptions of power supply or network connectivity shall not require manual user intervention once normal operating conditions have been restored.

Particular emphasis is placed on robust startup behavior, automatic recovery from temporary failures, and continued operation under varying environmental conditions. Future implementations may incorporate additional reliability mechanisms such as watchdog supervision, automatic recovery procedures, and redundant communication paths where appropriate.

Security

Monitoring information may contain sensitive data regarding the location and condition of a vehicle or vessel. Appropriate measures should therefore be implemented to protect both stored and transmitted information.

Communication between the gateway, backend services, and user interfaces should utilize authenticated and encrypted communication channels where appropriate. Access to monitoring information shall be restricted to authorized users through suitable authentication and access-control mechanisms.

The security architecture should be designed to accommodate future enhancements, including credential management, device registration, and secure remote configuration.

Roadmap

Phase 1 – Core Monitoring Prototype

The initial prototype focuses on validating the overall system architecture and communication concept. Key objectives include:

- Battery voltage monitoring
- Temperature and humidity monitoring
- Water ingress detection
- Smoke alarm integration
- Wi-Fi communication
- Backend connectivity and data visualization

- Historical data storage
- Event-based alarm handling

Phase 2 – Enhanced Connectivity

Following validation of the core functionality, additional communication capabilities may be introduced:

- Cellular communication (LTE/NB-IoT)
- Improved alarm delivery mechanisms
- Operation independent of local Wi-Fi infrastructure
- Enhanced gateway configuration and device management
- User-configurable alarm thresholds
- Simplified device onboarding and network configuration

Particular emphasis will be placed on minimizing installation complexity and enabling configuration without requiring direct access to the device software.

Phase 3 – Extended Monitoring Capabilities

Future versions may integrate data from existing onboard devices such as battery management systems, solar charge controllers, and navigation equipment.

- GPS positioning, tracking, and geofencing
- Integration of existing monitoring systems via Bluetooth (e.g., battery management systems, solar charge controllers)
- Detection of external power availability (shore power / auxiliary power connected or disconnected)
- Additional environmental sensors and monitoring functions

Phase 4 – Productization

Following successful validation of the technical concept, future development may focus on transforming the prototype into a deployable product.

Potential objectives include:

- Product-specific hardware design
- Optimized enclosure for marine and RV environments
- Simplified installation and configuration procedures such as plug and play features
- Remote firmware updates
- Configuration and data visualization app
- Improved reliability and diagnostics
- Compliance with applicable regulatory requirements
- Product documentation and user guides
- Integration with commercial notification and cloud services

This phase aims to improve usability, reliability, and maintainability while preparing the system for deployment beyond prototype and demonstration environments.

Example Use Case

Consider a sailing yacht moored in a marina for several weeks while the owner is away from the vessel. During this period, the yacht remains connected to the marina's Wi-Fi network, allowing the RV Monitoring Gateway to maintain communication with the backend service.

Every few minutes, the gateway collects information from its connected sensors and transmits the data to the backend. The owner can remotely access current and historical information such as battery voltage, cabin temperature, humidity levels, and the status of connected alarm inputs through a web or mobile interface.

After several days of heavy rainfall, a leak develops in a deck fitting and water begins to accumulate in the bilge. The water ingress sensor connected to the gateway detects the condition and immediately generates an alarm event. The event is transmitted to the backend and a notification is sent to the owner.

The owner receives the alert on their mobile device and can verify that the event is genuine by reviewing the current sensor readings. A local marina service can then be contacted before significant damage occurs.

Similarly, the system may notify the owner of unusually low battery voltage, excessive humidity levels that could lead to mold growth, or the activation of a connected smoke alarm. In each case, the RV Monitoring Gateway provides early awareness of developing issues while the vessel remains unattended.

Conclusion

The RV Monitoring Gateway presents a concept for a lightweight and modular remote monitoring platform for recreational vehicles and vessels. The proposed architecture focuses on providing remote access to essential status information while minimizing installation effort, power consumption, and dependency on proprietary ecosystems.

The concept described in this whitepaper serves as the foundation for the development of an initial prototype based on an ESP32 platform. Future work will focus on validating the communication architecture, evaluating power consumption, implementing reliable alarm handling, and assessing the feasibility of long-term unattended operation in real-world RV and marine environments.

The long-term objective is to establish a scalable and vendor-independent monitoring platform that can evolve from a proof of concept into a practical solution for recreational vehicle and boat owners seeking reliable remote monitoring capabilities.